

NagiosQL 3.5 — Installation Guide

Debian Trixie (13) · Nagios 4 via apt · Apache2 + PHP 8.4

1. Prerequisites

Component	Min version	Debian package
Apache2	2.4	apache2
PHP	8.2+	libapache2-mod-php8.4
MySQL/MariaDB	5.7+	mariadb-server or external
Nagios	4.x	nagios4-core
PHP modules	—	php8.4-mysql php8.4-gd php8.4-mbstring php8.4-curl php8.4-xml php8.4-zip

Install all packages:

```
apt-get update
apt-get install -y \
    nagios4-common nagios4-core monitoring-plugins nagios4-cgi \
    apache2 libapache2-mod-php8.4 \
    php8.4-mysql php8.4-gd php8.4-mbstring php8.4-curl php8.4-xml php8.4-zip \
    mariadb-server
```

2. Debian nagios4 package — default paths

Resource	Path
Main config	/etc/nagios4/nagios.cfg
CGI config	/etc/nagios4/cgi.cfg
Resource file	/etc/nagios4/resource.cfg

Resource	Path
Plugins	/usr/lib/nagios/plugins/
CGI binaries	/usr/lib/cgi-bin/nagios4/
Static files	/usr/share/nagios4/htdocs/
Daemon binary	/usr/sbin/nagios4
Var / runtime	/var/lib/nagios4/
External command pipe	/var/lib/nagios4/rw/nagios.cmd
PID file	/run/nagios4/nagios4.pid
Logs	/var/log/nagios4/nagios.log
Object cache	/var/cache/nagios4/objects.cache

3. NagiosQL directory structure

NagiosQL writes Nagios object files to `/etc/nagiosql/` — a directory separate from `/etc/nagios4/`.

```
mkdir -p \
  /etc/nagiosql/hosts \
  /etc/nagiosql/services \
  /etc/nagiosql/backup/hosts \
  /etc/nagiosql/backup/services
```

3.1 Permissions

Apache (`www-data`) needs write access; the Nagios daemon (`nagios`) needs read access:

```
chown -R www-data:nagios /etc/nagiosql
find /etc/nagiosql -type d -exec chmod 750 {} \;
find /etc/nagiosql -type f -exec chmod 640 {} \;
```

NagiosQL also needs write access to two core Nagios files:

```
chown www-data:nagios /etc/nagios4/nagios.cfg /etc/nagios4/cgi.cfg
chmod 640 /etc/nagios4/nagios.cfg /etc/nagios4/cgi.cfg
```

4. Apache2 configuration

4.1 Enable modules

`libapache2-mod-php8.4` requires `mpm_prefork` (`mod_php` is not thread-safe). The `nagios4-cgi` package installs `/etc/apache2/conf-available/nagios4-cgi.conf` with `Require ip` (private IPs only) and Digest Auth — disable it and use a custom `VirtualHost`:

```
a2dismod mpm_event
a2enmod mpm_prefork php8.4 cgi
a2disconf nagios4-cgi
```

4.2 VirtualHost — Nagios Core (port 80)

Create `/etc/apache2/sites-available/nagios4.conf`:

```
<VirtualHost *:80>
    RedirectMatch ^/$ /cgi-bin/nagios4/tac.cgi

    Alias /nagios4 /usr/share/nagios4/htdocs
    <Directory /usr/share/nagios4/htdocs>
        Options FollowSymLinks
        AllowOverride None
        AuthName "Nagios Access"
        AuthType Basic
        AuthUserFile /etc/nagios4/htpasswd.users
        Require valid-user
    </Directory>

    ScriptAlias /cgi-bin/nagios4 /usr/lib/cgi-bin/nagios4
    <Directory /usr/lib/cgi-bin/nagios4>
        Options ExecCGI
        AllowOverride None
        SetEnv NAGIOS_CGI_CONFIG /etc/nagios4/cgi.cfg
        AuthName "Nagios Access"
        AuthType Basic
        AuthUserFile /etc/nagios4/htpasswd.users
        Require valid-user
    </Directory>
</VirtualHost>
```

4.3 VirtualHost — NagiosQL (port 8081)

Create `/etc/apache2/sites-available/nagiosql.conf`:

```
Listen 8081

<VirtualHost *:8081>
    DocumentRoot /var/www/nagiosql
    DirectoryIndex index.php

    <Directory /var/www/nagiosql>
        Options FollowSymLinks
        AllowOverride None
        Require all granted
    </Directory>

    <Directory /var/www/nagiosql/config>
        Require all denied
    </Directory>
</VirtualHost>
```

Enable sites:

```
a2dissite 000-default
a2ensite nagios4 nagiosql
systemctl restart apache2
```

4.4 PHP — required settings

Create `/etc/php/8.4/apache2/conf.d/99-nagiosql.ini`:

```
[Date]
date.timezone = America/Sao_Paulo

[Session]
session.auto_start = 0

[File]
file_uploads = On
```

5. Install NagiosQL

Copy files to `/var/www/nagiosql/` :

```
cd /opt
tar xzf nagiosql_350.tar.gz
mv nagiosql35 /var/www/nagiosql
chown -R www-data:nagios /var/www/nagiosql
chmod -R 755 /var/www/nagiosql
chmod 750 /var/www/nagiosql/config
```

Run the installation wizard:

```
http://server/nagiosql/install/index.php    (port 80)
http://server:8081/install/index.php      (port 8081)
```

6. NagiosQL configuration (Administration → Config targets)

After installation, go to **Administration** → **Config targets** → **localhost** and set:

Field	Value
Base directory	<code>/etc/nagiosql/</code>
Host directory	<code>/etc/nagiosql/hosts/</code>
Service directory	<code>/etc/nagiosql/services/</code>
Backup directory	<code>/etc/nagiosql/backup/</code>
Host backup dir	<code>/etc/nagiosql/backup/hosts/</code>
Service backup dir	<code>/etc/nagiosql/backup/services/</code>
Nagios base dir	<code>/etc/nagios4/</code>
Import directory	<code>/etc/nagios4/conf.d/</code>
Picture directory	<code>/usr/share/nagios4/htdocs/images/logos/</code>
Nagios command file	<code>/var/lib/nagios4/rw/reload.trigger</code>
Nagios binary	<code>/usr/sbin/nagios4</code>

Field	Value
Nagios process file	/run/nagios4/nagios4.pid
Nagios config file	/etc/nagios4/nagios.cfg
CGI config file	/etc/nagios4/cgi.cfg
Resource file	/etc/nagios4/resource.cfg
Nagios version	4

Note — command file: NagiosQL uses a file-based reload trigger (`reload.trigger`) rather than the traditional external command pipe (`nagios.cmd`). The reload watcher script polls for this file and runs `nagios4 -v && systemctl reload nagios4` when it appears.

7. nagios.cfg — critical changes for NagiosQL

The Debian `nagios4` package ships default object files under `/etc/nagios4/objects/` that **duplicate** the definitions NagiosQL generates. Loading both causes `duplicate definition` errors that prevent the daemon from starting.

7.1 Comment out conflicting default entries

```
sed -i \
    -e 's|^cfg_dir=/etc/nagios-plugins/config|#cfg_dir=/etc/nagios-plugins/config|' \
    -e
's|^cfg_file=/etc/nagios4/objects/commands\.cfg|#cfg_file=/etc/nagios4/objects/commands.cfg|' \
    -e
's|^cfg_file=/etc/nagios4/objects/timeperiods\.cfg|#cfg_file=/etc/nagios4/objects/timeperiods.cfg|' \
    -e
's|^cfg_file=/etc/nagios4/objects/templates\.cfg|#cfg_file=/etc/nagios4/objects/templates.cfg|' \
    -e
's|^cfg_file=/etc/nagios4/objects/contacts\.cfg|#cfg_file=/etc/nagios4/objects/contacts.cfg|' \
    -e
's|^cfg_file=/etc/nagios4/objects/localhost\.cfg|#cfg_file=/etc/nagios4/objects/localhost.cfg|'
```

```
host.cfg|' \
/etc/nagios4/nagios.cfg
```

Files that conflict and why:

Default file	Conflict
objects/commands.cfg	NagiosQL generates commands.cfg in /etc/nagiosql/
objects/timeperiods.cfg	NagiosQL generates timeperiods.cfg in /etc/nagiosql/
objects/templates.cfg	NagiosQL generates host/service templates
objects/contacts.cfg	NagiosQL generates contacts.cfg + contactgroups.cfg
objects/localhost.cfg	NagiosQL manages all host definitions

7.2 Add NagiosQL cfg entries

Append to /etc/nagios4/nagios.cfg (idempotent — only if not already present):

```
grep -q "etc/nagiosql/hosts" /etc/nagios4/nagios.cfg || cat >>
/etc/nagios4/nagios.cfg << 'EOF'

# Configurations managed by NagiosQL (/etc/nagiosql/)
cfg_file=/etc/nagiosql/timeperiods.cfg
cfg_file=/etc/nagiosql/commands.cfg
cfg_file=/etc/nagiosql/contacts.cfg
cfg_file=/etc/nagiosql/contactgroups.cfg
cfg_file=/etc/nagiosql/contacttemplates.cfg
cfg_dir=/etc/nagiosql/hosts
cfg_file=/etc/nagiosql/hosttemplates.cfg
cfg_file=/etc/nagiosql/hostgroups.cfg
cfg_file=/etc/nagiosql/hostextinfo.cfg
cfg_file=/etc/nagiosql/hostescalations.cfg
cfg_file=/etc/nagiosql/hostdependencies.cfg
cfg_dir=/etc/nagiosql/services
cfg_file=/etc/nagiosql/servicetemplates.cfg
cfg_file=/etc/nagiosql/servicegroups.cfg
cfg_file=/etc/nagiosql/serviceextinfo.cfg
cfg_file=/etc/nagiosql/serviceescalations.cfg
cfg_file=/etc/nagiosql/servicedependencies.cfg
EOF
```

Also ensure `check_external_commands=1` is set:

```
grep check_external_commands /etc/nagios4/nagios.cfg
# check_external_commands=1
```

8. Runtime directories and reload trigger

8.1 `rw/` directory permissions

The `rw/` directory must be writable by both the Nagios daemon (`nagios`) and Apache (`www-data`). Use SGID so new files inherit the group:

```
mkdir -p /var/lib/nagios4/rw /var/lib/nagios4/spool/checkresults
chown nagios:www-data /var/lib/nagios4/rw
chmod 2775 /var/lib/nagios4/rw          # SGID: new files inherit www-data group

chown nagios:www-data /var/lib/nagios4/spool/checkresults
chmod 775 /var/lib/nagios4/spool/checkresults
```

8.2 `reload.trigger`

NagiosQL checks `file_exists()` on the command file before writing. Pre-create the trigger file so NagiosQL can write to it even before the first reload:

```
touch /var/lib/nagios4/rw/reload.trigger
chown nagios:www-data /var/lib/nagios4/rw/reload.trigger
chmod 660 /var/lib/nagios4/rw/reload.trigger
```

8.3 Validate and reload

```
/usr/sbin/nagios4 -v /etc/nagios4/nagios.cfg
systemctl reload nagios4
```

9. `cgi.cfg` — authentication mode

The Debian package sets `use_authentication=0` in `/etc/nagios4/cgi.cfg` by default. With Apache Basic Auth in front, this is correct: Apache handles authentication and every authenticated user gets full access to the Nagios CGI. Verify:

```
grep use_authentication /etc/nagios4/cgi.cfg
# use_authentication=0
```

Also create the stylesheets directory referenced in the VirtualHost:

```
mkdir -p /etc/nagios4/stylesheets
```

10. htpasswd — Nagios Core user

```
# Create the file with the nagiosadmin user
htpasswd -c /etc/nagios4/htpasswd.users nagiosadmin

# Fix permissions (Apache needs read access)
chown nagios:www-data /etc/nagios4/htpasswd.users
chmod 640 /etc/nagios4/htpasswd.users
```

11. Final verification

```
http://server:80/cgi-bin/nagios4/tac.cgi    → Nagios Core (Basic Auth)
http://server:8081/                        → NagiosQL (own login)
```

In NagiosQL: **Tools** → **Support page** shows the status of all configured paths.

Validate the Nagios config and confirm zero errors:

```
/usr/sbin/nagios4 -v /etc/nagios4/nagios.cfg 2>&1 | grep "Total Errors"
# Total Errors:    0
```

Appendix — Differences from older guides (nagios3 / pre-2020)

Old guide	Debian nagios4 / trixie
/etc/nagios/	/etc/nagios4/
/opt/nagios/bin/nagios	/usr/sbin/nagios4
/var/nagios/	/var/lib/nagios4/
/var/nagios/rw/nagios.cmd	/var/lib/nagios4/rw/nagios.cmd
/var/nagios/nagios.lock	/run/nagios4/nagios4.pid
/usr/local/nagios/libexec/	/usr/lib/nagios/plugins/
PHP 5.x / 7.x	PHP 8.4
php5-mysql	php8.4-mysql
Apache wwwrun	Apache www-data
nagios.cmd as command file	reload.trigger (file-based reload)
Default object files loaded	Default object files must be commented out